

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, Systems Testing & Validation Aug 2025 to May 2026

- Validated **14 interoperability scenarios** across networking, dependency, and **failure-recovery** paths to confirm recovery behavior before large experiment runs.
- Automated provisioning, configuration, and **telemetry** for containerized test clusters, cutting experiment setup effort **55%** and making every sweep reproducible from config.
- Reduced rebuffering **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configs.
- Mentored **200+ students** across **3** courses (Programming Paradigms, AI, ML) through debugging sessions, code reviews, and grading.

Turito | Software Engineering Intern, Reliability & Performance Jun 2022 to Oct 2022

- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Reduced production diagnosis time **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Slashed p95 **CPU 22%** by profiling high-frequency media paths and eliminating excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Debugged **ICE/STUN/TURN** connection setup and degraded-network recovery across a **WebRTC** control plane and **RTP/RTCP** media paths spanning about **12 regions**.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Cut performance-investigation time about **40%** by building tooling on Rust workers, **Python** aggregation, **SQLite**, Flask, and Chart.js.

TECHNICAL SKILLS

Languages: Python, C++17, Rust, Java, SQL, Bash

Test & Validation: test automation frameworks, pytest, Locust load testing, regression suites, interoperability and failure-recovery scenarios, failure classification, root-cause analysis

Systems & Tooling: Linux, Docker, Git, CI/CD, raw-socket networking, CPU and lock-contention profiling, telemetry pipelines, structured logging

Data & Analysis: NumPy, Pandas, SQLite, statistical significance testing, dashboards, reproducible experiment configs

PROJECTS

TCP Sliding Window Simulator | *Python, raw sockets, Linux*

- Implemented handshake, sliding window, cumulative ACKs, and retransmission over raw sockets with live packet-state visualization.

Proxima, Vector Search Engine | *C++17, AVX2/SIMD, HNSW, pybind11*

- Reached **0.999 recall** at **12,049 QPS** single-threaded, **1.8x** hnswlib and **2.5x** FAISS, with an HNSW engine written from scratch in C++17.

OPEN SOURCE

- fission/fission:** 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration
- corsairdev/corsair:** 6 merged **webhook signature verification** fixes (Telegram, Teams, SharePoint, Monday, GitLab, Jira) that fail closed

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Computer Networks, Distributed Computing, Advanced Algorithms, Database Systems, Machine Learning, Artificial Intelligence

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)